RAHA: how to pitch it

Team Antimatter · SRM Institute of Science and Technology (SRMIST), Chennai · Live simulation: (add the live URL after deploying)

Team Antimatter · PS2 "The Stranded Network" · SRMIST Chennai

This page has five parts:

1. The full story: problem, root causes, solution, constraints, business
2. A five-minute script that walks through the simulation chapter by chapter
3. The judges' Q&A
4. The honest limits to state yourselves
5. Which numbers to quote

Every number below is either a cited fact (see `data/scenarios.json`) or produced by the simulation (see `results/summary.json`). Don't quote numbers from the old PPT that aren't here.

1. The story in one breath

When a train line fails, thousands of people are stranded at once. Everyone opens a cab app, the fare surges, forty people heading the same way book forty cabs, the replacement buses arrive late and in the wrong place, and the crowd crushes at the doors. RAHA turns that free-for-all into a coordinated recovery:
• bridging buses with passes that match seats exactly
• the most vulnerable and most urgent riders first
• Koottu shared cabs, so sharing costs less than riding alone even at surge
• a geofence that pulls in bike taxis and delivery partners
It keeps working with the internet cut, and when a data feed changes shape. We prove it on five real disruptions, three in India and two abroad, rebuilt from real map data.

2. The problem (PS2: The Stranded Network)

What happens when a rail line stops, and why. Each cause has a source in the sim (press **S**).

Real incident	What went wrong	Why it happens (cause type)
Chennai (Tambaram), 28 Feb 2026	Signal failure. EMUs halted over an hour. People walked the tracks. Promised special buses didn't come.	**Space:** footfall about 1.5 lakh a day, infrastructure lagging. **Time:** services already cut from 204 to 164. **System:** almost nobody keeps bus reserves for bridging (Pender et al., 2013).
New Delhi, 15 Feb 2025	18 died on FOB-3 stairs. 15 trains were delayed between 20:30 and 22:30, and about 2,600 extra general tickets were sold in two hours.	**Time:** tickets are not linked to seats. **Information:** two trains had near-identical names. **Space:** one staircase between platforms.
Mumbai (Kurla), 19 Aug 2025	Rain suspended CSMT–Thane. Thousands were stranded.	**System:** drains handle about 25 mm/h, and only 3 of 45 outfalls have floodgates. **Space:** a 2.4 m footbridge at Elphinstone killed 23 people in 2017.
Singapore (Jurong East), Sep 2024	Six days without trains. Bridging buses ramped from 39 to 70. 30-minute queues.	**System:** maintenance lapse (SMRT fined S$2.4m). **Space:** bridging point an 8-minute walk away. **Information:** riders confused between shuttle, bridging and regular buses.
Madrid (Atocha), 28 Apr 2025	Grid collapse at 12:33. Metro evacuated. About 35,000 rail passengers stranded. Mobile networks degraded.	**System:** voltage-control failure (ENTSO-E). Every cloud-dependent app stopped at once.

Frame it with Fruin's FIST model. Crowd disasters come from Force, Information, Space and Time, not from "panic". Add two causes of our own:

- **System:** nobody plans the bridging.

- **Market:** surge pricing, and solo booking.

3. The solution: RAHA

One recovery engine with four tools. Every one of them is modelled in the simulation.

1. **Bridging buses with capacity-exact passes.**
   - Riders are grouped by real destination corridor.
   - Buses go where the most urgent riders wait, and ramps go where wheelchair users are.
   - A pass exists only if a seat exists, so nobody is turned away at a door.
   - Pass holders are released in waves from a holding area, which keeps crowd density at level D (comfortable) or better.
2. **Fair priority, explained.**
   - Score = 35·V (vulnerability) + 30·D (deadline urgency) + 25·W (wait, which only ever rises) – 10·C (detour).
   - **Willingness to pay is never an input.**
   - A rider who misses a seat gets a **refusal proof**: their score, the cutoff score, their rank, and a SHA-256 record hash.
3. **Koottu shared cabs** (the lead's app idea, modelled in the sim).
   - Surge is charged **per car, not per person**. Four people at the 2× cap each pay half the normal fare. Three people pay about two-thirds.
   - Pitch line: **"sharing costs less than riding alone at the normal fare."**
   - Groups form only inside **trusted circles** (Zimride, Cornell, which became Lyft).
   - Four phones get four quotes, and **the cheapest books**.
   - The split uses the **Shapley value** (airport game). Then the "never more than alone" rule is **checked directly** and anyone who would break it is dropped. The guarantee does not rest on Shapley alone.
   - The fare is checked against the **2× legal cap** (Motor Vehicle Aggregator Guidelines 2025). Outside India, the sim says "cannot check".
   - Pooling also means fewer booking requests, so surge rises less.
4. **Geofence + two-wheelers + essentials** (the Swiggy/Uber-style layer).
   - Riders, cabs and delivery partners inside the geofence are alerted.
   - Bike-taxi and delivery partners take urgent solo riders, **where state law allows**. Karnataka banned bike taxis in 2025.
   - Riders waiting more than 45 minutes are flagged for water and food drops. Cyclone Michaung showed that stranded riders go hungry and thirsty.

4. The two constraint cards (and how RAHA adapts)

Card	What it says	How RAHA adapts	Where the sim proves it
Constraint 1	No internet or third-party APIs. The essential journey must still work locally.	The station edge node on UPS power runs grouping, scoring, allocation and pass signing. Handhelds verify passes offline. Kiosks and staff tokens serve riders with no signal. Koottu books if any one phone in the group has signal.	**Madrid blackout scenario:** at 12:33 a cloud-only coordinator collapses into free-for-all while RAHA keeps allocating. A 200-seed sweep covers phone reach of 100%, 70%, 30% and 0%. The simulator itself is offline: the **EXTERNAL REQUESTS 0** badge counts and blocks any outside call.

Card	What it says	How RAHA adapts	Where the sim proves it
Constraint 3	An API changes its response: a field is renamed, nested, retyped or becomes optional.	One adapter sits in front of every feed. It resolves aliases, coerces types, falls back to last-known-good values and a local vehicle registry, and runs invariant checks for **semantic drift**: the MTA changed bus feeds from 0-based to 1-based stop numbering in May 2025. It fails safe: no trusted seat count means no passes.	**Singapore line-closure scenario:** pick any of six mutations live, and the JSON panel shows what the naive parser reads versus the adapter. A 200-seed table compares empty departures, turned away and worst wait.

5. How we prove it: the simulation

- **Real data.** The five islands are built from OpenStreetMap. Timelines and parameters come from cited sources: the Indian Railways and Chennai MTC GTFS feeds, Census 2011, LTA, ENTSO-E and others. Every input carries a tier from T1 (observed) to T4 (assumption, stress-tested). Press **D**.
- **One engine** (`src/sim/engine.js`) drives both the 3D replay and the statistics, so the numbers on screen are the numbers in the results file.
- **Honest rivals.** The same seed is run under each of these:
  - free-for-all (status quo)
  - first-come-first-served app
  - market pricing
  - pure throughput
  - RAHA
- **Sanity bound.** Clearance can't beat (waves – 1) × the shortest round trip.
- **Sensitivity.** Each result is re-run with door crush at 0, demand halved and doubled, and the fleet halved.

6. Business: who pays, and is it viable?

- **Riders pay nothing.** The Koottu side is free, and adoption comes from saving money during surge.
- **Operators pay (B2G).** Per-station annual licence plus incident drills.
  - Indian Railways allotted about ₹12,018 crore for passenger amenities in 2025-26, and is adding holding areas and AI monitoring at **76 stations**.
  - RAHA is the capacity-linked release and bridging layer for exactly those holding areas.
  - The simulator doubles as the planning tool that sizes bridging reserves.
- **Campuses and employers (B2B circles).** Per-member monthly fee for guaranteed rides home for students and shift workers when the network fails. This is the Zimride model.
- **Delivery and bike-taxi platforms.** Per geofenced dispatch lead: paid trips for idle partners exactly when demand spikes.
- **Events** (Kumbh, stadiums). Per-event deployments.
- **Proof the market pays:**
  - **Via** (transit software) listed on the NYSE in 2025 at about **$3.5 bn**, with **over 90% of revenue from government agencies**.
  - **Chalo** works with TNSTC, DTC and KSRTC.
  - One railway division's information-board tender alone is about **₹2.13 crore**.

- **Go to market.** Pilot at Chennai (Tambaram station) with Southern Railway Chennai Division and MTC through SRMIST (real riders, a real corridor next door), then the 76 holding-area stations.
- **Risks.** Long public-sector sales cycles, state rules on two-wheelers and multi-stop pooling, and dependence on riders entering fares (we never scrape).

The **Business case** chapter in the sim computes the value per incident from the measured runs:

- rider-hours saved
- turned-away attempts avoided
- cab trips avoided
- money saved by pooled riders
- the change in deadlines met

7. Five-minute script (presenter at the clicker; → advances)

Time	Chapter	Say
0:00	Title → Globe	"Five real disruptions, three in India, two abroad. Every island you'll see is built from real map data."
0:20	The 2 constraint cards	"Our cards: no internet, and an API that changes shape. Both are built into the product, and you'll watch both fail and recover."
0:45	Chennai: What happened	"Twenty-eighth of February this year, 8:50 am. A signal fails south of Tambaram. People walk the tracks. The promised buses don't come."
1:15	Why it happened	"Fruin's four lenses: Information, Space, Time, Force. Plus the system cause: nobody plans the bridging."
1:40	Replay twice	"Same people, same buses, same seed. Left: as it happened. Right: RAHA." Point at **turned away**, **door crowd level F → D**, **riders per cab**.
2:20	RAHA up close	"Every pass is a real seat. A wheelchair user is first in line. Koottu: the cheapest of the group's phone quotes books, the Shapley split is checked against each solo fare, and the surge is checked against the legal 2× cap." Point at the refusal proof.
3:00	Madrid: Constraint 1	"12:33, Madrid loses power and mobile data. The cloud coordinator falls back to chaos. RAHA's edge node keeps going. And this page itself makes zero external requests."
3:40	Singapore: Constraint 3	"Mid-incident the bus feed renames its capacity field. The naive parser sends buses out empty; our adapter doesn't." Click **semantic**: "Same field name, new meaning, caught by an invariant."
4:20	Finale → Business case	"One engine, five real cities, measured over 200 runs each. Free for riders, paid by the operators who already fund holding areas at 76 stations."
4:50	Close	"Stranded networks will keep happening. RAHA makes sure nobody is left at the door."

If time is short, skip New Delhi and Kurla: press **5** for Madrid (Constraint 1), then **4** for Singapore (Constraint 3).

When a judge asks "does it really work offline / with a changed API?"

1. Press **J**. The Judge check runs every constraint test live in the browser.
2. Switch Wi-Fi off and reload the deployed site. The service worker serves everything from cache.
3. Press **J** again. Everything still passes, and network status now reads "offline".

8. Judges' Q&A

Question	Answer
Is this real data?	The geography, timelines, timetables and published counts are real. Individual riders are sampled from real distributions, because no per-person data exists and it would be a privacy problem. Press **D** to see the tier of every input.
Where does the 3.4 riders per cab come from?	The engine, over 200 seeds, in the Chennai scenario. Groups are limited by trusted circle, corridor and each rider's limit, so it's below 4.
Is Shapley enough to guarantee "never more than alone"?	Not on its own once detours appear. We check every share against that rider's solo fare and drop anyone over it. The guarantee is enforced in code, not assumed.
What if nobody in a group has signal?	Grouping, scoring and passes still work on the edge node. Booking a ride-hail cab needs one phone with signal. Without any, riders fall back to buses and two-wheelers that are dispatched locally.
Your 45-minute evacuation claim?	We don't make it. The sim checks a physical lower bound and reports measured clearance with confidence intervals.
Bike taxis are banned in some states.	The two-wheeler layer is off where it isn't legal (Karnataka 2025). It's a modelled option, not a promise.
Why would Railways buy this?	They are already funding holding areas and AI monitoring at 76 stations. RAHA is the logic that decides who leaves the holding area, when, and on what.

9. Honest limits (say them yourselves)

- RAHA can't switch surge off. It makes surge hurt less.
- Rider populations are synthetic, drawn from real distributions. Assumptions (T4) are shown and stress-tested.
- The legal 2× cap is a central guideline. No Tamil Nadu notification adopting it was found, so the app says "cannot check" without a verified base fare.
- Grouping is a heuristic, not an optimum.
- Multi-stop pooling may need permits (Karnataka halted pooling once).
- The old PPT's numbers (–68%, 3.8×) are superseded by the measured numbers in the sim.

10. Numbers to quote

After `node scripts/run_experiments.mjs 200`, take them from `results/summary.json` or from the **Measured results** chapter.

The headline format:

> "Across 200 simulated runs of the real 28 February 2026 Tambaram failure, the worst-off rider waited X minutes less than under free-for-all and Y less than a first-come-first-served app, with zero people turned away at doors and Z riders per shared cab."